

Java Data Structures

Data Structures

Primitive Types:

These types store simple values directly in memory and have high performance. These types are divided into numeric and non-numeric categories.

Reference Types:

These types store the address (reference) of an object in memory. Their default value is null.

Primitive Integer Types

byte

- Size: 8 bits
- Range: -128 to 127

short

- Size: 16 bits
- Range: -32,768 to 32,767

int

- Size: 32 bits
- Range: -2,147,483,648 to 2,147,483,647

long

- Size: 64 bits
- Range: -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807

Primitive Floating-Point Types

float

- Size: 32 bits
- Range: approx. $1.4\text{e-}45$ to $3.4\text{e+}38$
- Description: Single-precision floating-point

double

- Size: 64 bits
- Range: approx. $4.9\text{e-}324$ to $1.8\text{e+}308$
- Description: Double-precision floating-point

Primitive Non-Numerical Types

char

- Size: 16 bits
- Range: '\u0000' to '\uffff' (0 to 65,535)
- Description: A single Unicode character

boolean

- Size: JVM-dependent (typically 1 byte or 1 bit)
- Values: true or false
- Description: Logical value (truth value)

Primitive Types Summary Table

Type	Size (bits)	Range / Values	Default Value
byte	8	-128 to 127	0
short	16	-32,768 to 32,767	0
int	32	-2,147,483,648 to 2,147,483,647	0
long	64	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	0L
float	32	approx. 1.4e-45 to 3.4e+38	0.0f
double	64	approx. 4.9e-324 to 1.8e+308	0.0d
char	16	'\u0000' to '\uffff' (0 to 65,535)	'\u0000'
boolean	JVM-dependent	true or false	false

Primitive Non-Numerical Types

char

- Size: 16 bits
- Range: '\u0000' to '\uffff' (0 to 65,535)
- Description: A single Unicode character

boolean

- Size: JVM-dependent (typically 1 byte or 1 bit)
- Values: true or false
- Description: Logical value (truth value)